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Knowledge informed hybrid machine learning in agricultural yield prediction

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ABSTRACT

Research on yield predictions is dominated by two approaches: machine learning and process-based models. Machine learning has shown impressive results in capturing complex relationships but is often limited by data availability in agriculture. Conversely, process-based models, with over 60 years of research history, simulate crop growth processes using biophysical equations. Here, we present a method to transfer domain knowledge from the Decision Support System for Agrotechnology Transfer framework (DSSAT) using the Nwheat crop simulation process-model into neural networks and random forest for predicting wheat yield at field scale. Expanding the feature and distribution space involved simulating crop parameters and synthetic samples through the utilization of observed and historical weather recordings, as well as future climate projections. We demonstrated that neural networks can learn both general crop growth and yield processes and then effectively adapt to regional, field-specific growth patterns using synthetic and high-resolution field data. This approach boosts overall performance and reduces model error by 8 % compared to a purely data-centric model without process-knowledge transfer and solely trained on observed field data and features. Synthetic samples generated from warmer conditions were the greatest driver for improvements and we showed that the climate scenario for data generation is more important than the actual synthetic data set size. The proposed method shows the potential of combining process-based and machine-learning models, highlighting the potential to leverage the strengths of both methods in a collaborative manner.

1. Introduction

Modeling the complex interaction between climate and environment, and the effects of farmers' management is crucial to gaining reliable forecasts of crop yield. Farmers around the world are challenged to produce higher yields and use less resources, all while climate change increases extreme weather events and makes crop growth less predictable (BrachoMujica et al., 2024). Forecasts of yield and assessments of influencing factors, especially the climate, can help farmers take appropriate management measures and harvest more stable yields with greater certainty and quantity (Weersink et al., 2018). Machine learning (ML) models can be helpful in these efforts, as they are very well suited to modeling non-linear, complex phenomenon and can be flexibly adapted to new crops or regions. A key component of successful training and application of ML models is the availability of data. The accuracy and reliability of data-driven models increase significantly when more

high-value data is available, and conversely the lack of data negatively impacts the performance of these models (Gordon et al., 2021).

In sectors where data acquisition is expensive or time-consuming, scarcity of data further intensifies this issue. Agriculture faces these problems as crop cycles for most crops contain only one harvest per year and thereby inherently limit the ability for data recording (Maestrini et al., 2022; Osinga et al., 2022). In addition, climate change induces increasingly extreme weather conditions that pose challenges for simulation using ML, especially when considering that these scenarios are at the tails of the weather distribution (Masson-Delmotte et al., 2021). Compounding this limitation are labor-intensive survey methods, missing incentives for farmers to contribute data, and the lack of suitable technology to record data in many parts of the world (Gackstetter et al., 2023). The great advantage that can be provided by ML has yet to be realized in agriculture (Weersink et al., 2018).

In other domains with data scarcity or where training in real-world is

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not possible (e.g. in robotics), combining ML with scientific laws or domain knowledge has become an area of active research. Examples can be found in many fields, including robotics (Karniadakis et al., 2021), climate modeling (Kashinath et al., 2021) or health care (Xu et al., 2023).

Process-based crop simulation models contain a long history of research regarding crop growth and bio-physical crop physiology but are difficult to parameterize, lack generalization capabilities, and often require intensive setup efforts (Lobell & Asseng, 2017; Seidel et al., 2018). On the contrary, ML models have the potential to be trained more accurately for specific situations, might discover patterns that are not accounted for in crop simulation models yet, and adapt better to local modalities (Maestrini et al., 2022). We present different strategies for how both model approaches could be linked to each other to predict wheat crop yield. The concept is shown in Fig. 1. Our integration approaches are based on generating synthetic yield data with the process-based crop simulation model Nwheat from the ‘Decision Support System for Agrotechnology’ (DSSAT-Nwheat), applied across a wide range of climate, soil, and management conditions. To generate the synthetic yield data, we use three weather data sets: The first uses observed weather data (OWD) from the time period of the field experiment recordings (2005–2021), the second historical weather data from the period 1990–2005 (HWD), and the third approach synthetic weather data from climate projections from 2037 to 2099 (CP).

Our research contribution contains three components: We demonstrated the considerable potential of climate projections to expand the range of weather conditions for synthetic data. Secondly, we employed a single reference crop to calibrate DSSAT-Nwheat and the learned rules of crop growth appeared to be highly generalizable for another 751 widely diverse cultivars. The third aspect involved applying a multi-loss function to softly constrain our neural network, requiring it to learn the entire crop growth pattern from DSSAT-Nwheat in addition to yield prediction. We apply our approaches to wheat, one of the most important crops for human nutrition, on a field-level scale where farmers make decisions. To do this, we take advantage of a unique dataset of over 75,000 observations from wheat experiments conducted throughout Germany in 2005–2021.

In this paper, we aim to integrate expert domain knowledge from process-based crop simulation models into ML models to combine the strengths of both approaches.

We set up the following hypotheses:

1. ML can learn fundamental structures of crop growth from process-based models and benefit in terms of performance
2. Major improvements in performance will be gained from synthetic data generated with climatic features around the tails of the distribution space
3. The use of synthetic data will be most beneficial in years and locations with extreme climatic conditions – in particular, hot and dry conditions which occur more frequently because of climate change

2. Related work

Predictions for crop yield using process-based models have a long research history going back to the early 1960s. These models are designed to simulate the physiological and bio-chemical processes that take place in crop fields (Wallach et al., 2018). They are capable of simulating a range of different crop variables like yield, biomass, leaf or root growth, and health conditions like drought or nutrition deficiencies (Boote et al., 2013). Thereby they contain a lot of science-based domain knowledge in general gained from field trials and contain fundamental growth functions, but many processes are still not fully understood (Jones et al., 2003; Maestrini et al., 2022). Additional limitations include that crop simulation models are very sensitive to their input data, difficult to calibrate and tend not to perform well on individual samples (Wallach et al., 2021). Numerous studies have been conducted for yield simulation of wheat using various crop simulation models on field level and show an average error around 13 %–17 %, and sometimes up to ~23 % (Palosuo et al., 2011; Attia et al., 2016; Yang et al., 2019).

ML-based approaches for yield prediction have become more popular in recent decades. They come with certain advantages, mainly the input vector is easy to modify and makes the integration of new data sources or multi-modal data sets simple, as well as the ability to be flexible with regards to spatial scale. Most ML approaches, with larger datasets and higher spatial and temporal variation, are usually derived from experiments conducted at national or county spatial scales, with fewer studies performed at the field level. Studies on national or county level wheat prediction show accuracies of ~0.4–1 t/ha RMSE (Wang et al., 2020; Cao et al., 2021; Srivastava et al., 2022). Leng & Hall (2020) benchmarked crop simulation systems against ML methods for corn yield prediction and found the latter outperform. Predictions on field level tend to show higher RMSE than on national or sub-national (county, state) level and vary between ~0.4–1.1 t/ha (Marszałek

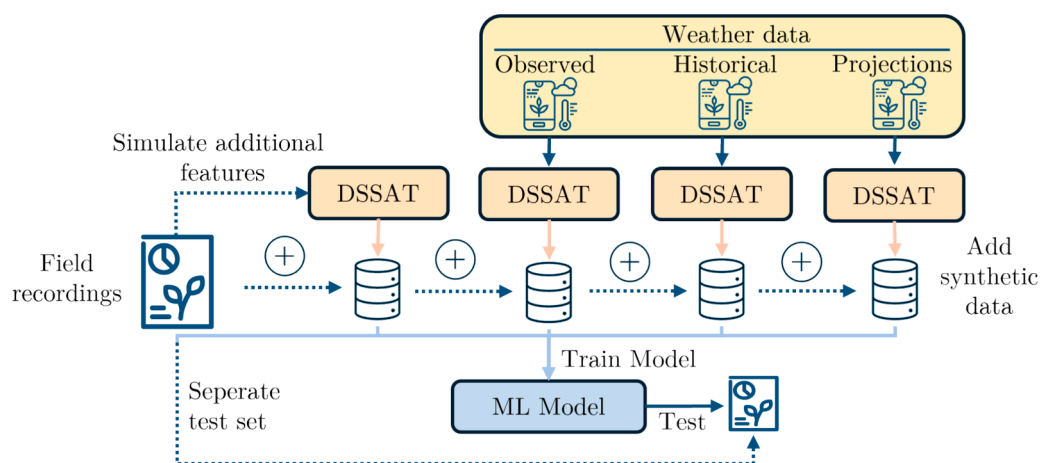


Fig. 1. Different data sets were generated by adding additional features or samples and extending the distribution space of the training sets. The training sets complement each other continuously. To generate the synthetic features and samples the process-based crop model Nwheat from the Decision Support System for Agrotechnology Transfer framework (DSSAT-Nwheat) was applied. Historical (1986–2004), observed (2005–2021), and future weather projections (2037–2099) were used for the DSSAT-Nwheat simulations and to extend the climatic feature distribution. For testing, one of the years was separated from the recorded field data set. This process was repeated iteratively over all years.

et al., 2022; Zhang et al., 2023) which is also reported when direct comparisons regarding the spatial scale are made (Cao et al., 2021). The data set sizes in mentioned national/subnational forecasting typically contain a couple of thousand data samples, often obtained from national statistic bureaus, with a history of up to a few decades. Field-level experiments often contain very few years (the mean was 2 years in the mentioned studies) and are typically geographically concentrated resulting in often less than 1,000 samples per data set. To our knowledge, very few authors have published crop yield prediction at the field scale with a longer time period (>10 years) and more than 50,000 field recordings (Deines et al., 2021).

Process and data-driven approaches are very different by method, but comparisons in the past have shown that both can be seen as reliable predictors, even though they can show different prediction behavior (Lobell & Asseng, 2017). Different ways of integration include the simulation of new synthetic features for ML approaches using process-based models (e.g. simulations or biomass, leaf growth, etc.). Feng et al. (2020, 2019); Corrales et al. (2022); Shahhosseini et al. (2021); Paudel et al. (2021), use ML to predict (missing) input data of process-based models (Saha et al., 2021), or use process-based models to generate synthetic samples to conduct transfer learning (Maestrini et al.,

2022; Kallenberg et al., 2023). The latter can also be applied to generate metamodels which refer to ML models trained to learn the crop simulation model on synthetic data (Xu et al., 2021; Lobell et al., 2015; Liu et al., 2024; Pylianidis et al., 2022). Reported improvements range from 7 % to 40 % across different crop types, with extreme climatic years benefiting the most (Shahhosseini et al., 2021; Kallenberg et al., 2023).

Using synthetic data for transfer learning is a common approach in computer vision, where virtual worlds and their game engines contain scientific and human-defined rules (Fabbri et al., 2021). However, across domains, several authors mention that a gap between simulation and reality remains, and real-world predictions often do not achieve model performance on simulated data (Heiden et al., 2020).

Adapting the loss function to constrain the learning with domain knowledge is a well researched approach in tasks where scientific laws and equations can be applied (Karniadakis et al., 2021). Several authors showed that constraining the learning behavior can improve the models in domains of robotics (Karniadakis et al., 2021), fluid dynamics (Sun et al., 2020) or weather modeling (Kashinath et al., 2021).

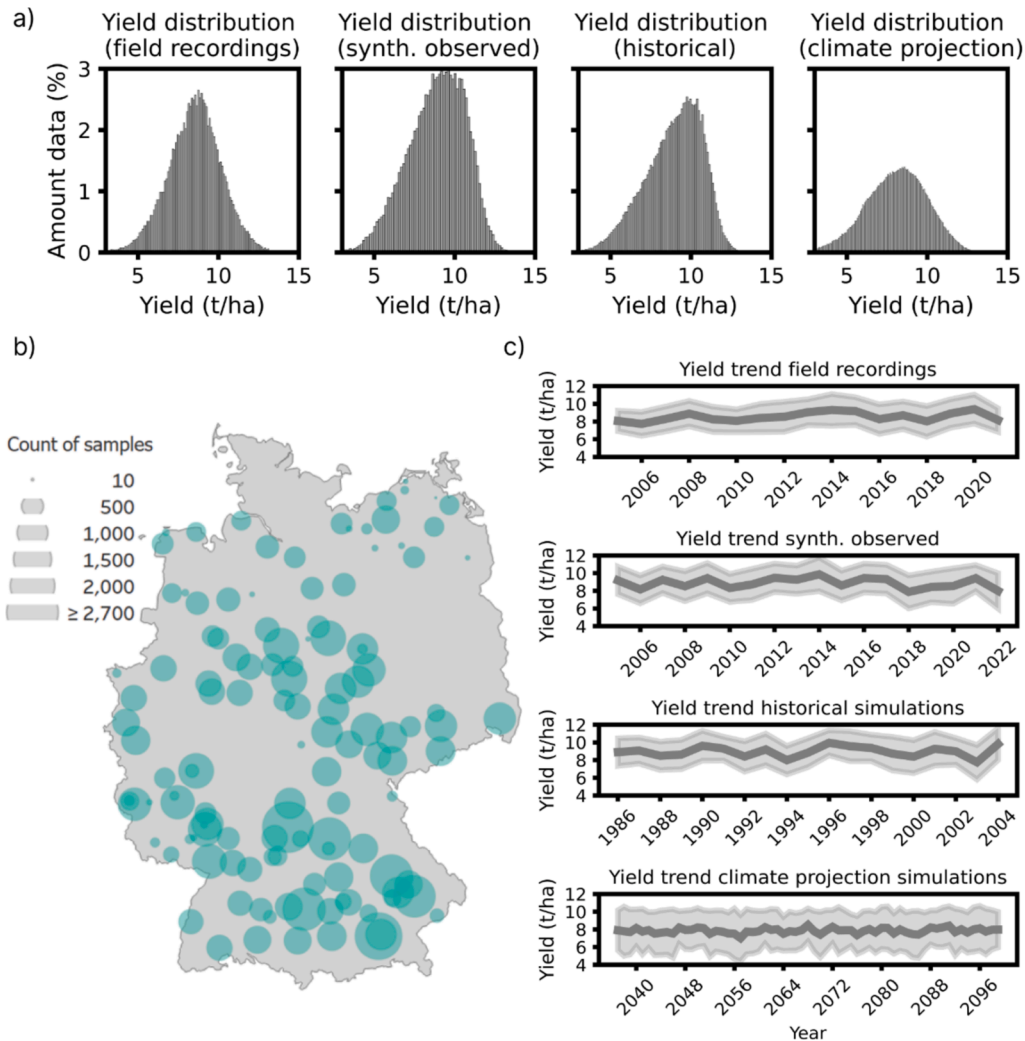


Fig. 2. a) Shows the yield distribution for the ground truth data set (field recordings) and the three synthetic data sets from historical (1986–2004), climate projection (2037–2099) simulations and simulations from weather data in the observed time period (2005–2021). b) Shows the spatial distribution of the sites across Germany. The bubble size indicates the length of the data set per site. c) Trendlines of the yield variable over the time periods in the data set. The grey filling represents the standard deviation.

3. Material and methods

Two distinct ML algorithms were employed for the experiments: a feed-forward neural network (NN) and a random forest (RF). These models were chosen for their versatility and ability to capture complex relationships within the data and are among the most used ML methods in related studies (see [section 2](#)).

3.1. Data

Our data set contains 76,466 recordings of wheat yield over 17 years (2005–2021), with 751 cultivars across 119 sites from seven states in Germany. The data were obtained from several different field experiments, mainly cultivar trials (where different cultivars are benchmarked each year) and fertilizer experiments. The recordings contain information about crop management (cultivar, fertilizer applications) and details about the site-specific geographic conditions (soil type, nutrition supply, etc.). For detailed information about the yield data and spatial distribution see [Fig. 2](#) and the [appendix](#), section A for feature description. Weather data (daily temperature, precipitation and solar radiation) were gathered from the German Weather Service ([DWD, 2024](#)), with the average weather station 8 km away from the nearest field trial. Three additional climate indices were included in the analysis: standardized temperature (1) and precipitation (2) index and growing degree days (3) calculated according to [McKee et al. \(1993\)](#) as detailed in [appendix](#) section 1.2. The standardized climate indices measure site-specific climate anomalies in temperature and precipitation. Cold or wet conditions are classified when the respective index falls below -1 , heat or drought when they exceed $+1$. Growing degree days quantify heat units required for crop development.

All weather data were aggregated monthly for use in ML models.

3.2. Neural network

The NN was implemented using a multi-layer perceptron architecture, including two layers with a hidden size of 128, and thereby kept rather small. Rectified Linear Units (ReLU) served as activation functions for the hidden layers, Adadelta with a learning rate scheduler as optimizer and the mean squared error loss (MSE) as default objective function. The training was interrupted when 25 epochs in a row no improvement on the validation set occurred. Results from NN are averaged over five repetitions with different random initialization. See [appendix](#) section 2.5 for details.

3.3. Random forest

The RF regressor was implemented from the sci-kit learn package ([Pedregosa et al., 2011](#)). Hyperparameter tuning did not lead to any significant changes so the default values were kept in place.

3.4. Baseline models

Two baseline models were established as benchmarks for the ML models. We argue that both – the process-based model and the site-specific expected value, the simplest model – must be outperformed to justify the effort of developing a hybrid ML approach. First, DSSAT-Nwheat (see [section 3.5](#)) was used to simulate the yield for each experiment. Second, the long term average yield for each site was used as a predictor ($E[x]$). The latter approach is considered as a robust and practically relevant method, as it mirrors the approach used by German farmers to estimate future yields and determine maximum fertilizer application rates in accordance with the European Union's Nitrates Directive. We refer to [appendix](#) section 2.5 for details of model settings.

3.5. Dssat-nwheat

For various scenarios, synthetic crop features and yields were simulated using the process-based model 'Nwheat' from the 'Decision Support System for Agrotechnology Transfer' (DSSAT-Nwheat, v4.7.5, [Jones et al., 2003](#)). The aim was to transfer the structural patterns of crop growth behavior into the ML models, by applying DSSAT-Nwheat on a reference crop. For example, the data-driven approaches should be trained in how crops react to drought conditions in general instead of learning it for every single cultivar.

DSSAT-Nwheat contains various input parameters that can be either approximated using statistical tools, offered by the DSSAT framework, or adjusted manually. Adjusting in the context of crop model setup refers to estimating the value of crop-relevant parameters (e.g. the number of kernels per ear, ears per plant, kernel weight of the specific cultivar), trying to find a reasonable constellation until the DSSAT-Nwheat model matches a set of yield observations. DSSAT-Nwheat was only parameterized for the cultivar 'JB Asano', one of the cultivars that appeared the most regarding years with recordings and sites that planted this crop. We adjusted the cultivar parameters manually, based on cultivar data we retrieved from the seed supplier, experimental records from our data set and experiments from [Röll et al. \(2020\)](#). This approach led to more accurate validation results compared to using the statistical estimators provided by the DSSAT framework. Additionally, several (regional) databases have been used to calculate the most realistic soil information input required by DSSAT-Nwheat ([BKG, 2020](#); [Boden, 2005](#); [BMUV, 2003](#)). We refer to [appendix](#) section 2.4 for the details of the DSSAT-Nwheat setup.

3.6. Pre-processing and leave-one-year-out cross validation

We conducted data pre-processing by one-hot-encoding of the categorical features and standardizing the continuous features to a mean of zero and a standard deviation of one.

To perform a realistic test scenario, we extracted one year from the set of 17 years and separated it as a test set. The remaining data set were split randomly into 80 % train and 20 % validation sets. Synthetic data were removed from the validation set to avoid overfitting on DSSAT-Nwheat simulations. When a convergence of the training was achieved, the model was tested once in the extracted test year. The whole process was then repeated across all 17 test years.

3.7. Synthetic crop data

DSSAT-Nwheat was used to simulate different features and samples, namely crop variables (e.g. leaf weight, root length) and conditions (e.g. drought stress). To generate synthetic yield samples, we combined randomly drawn management regimes from the crop experiments, soil conditions from the different sites, and daily weather data from either recorded (observed, historical) or synthetic (climate projection) sources.

3.8. Synthetic weather data

We have created a set of synthetic weather data using climate projections for the years 2037–2099. The climate simulations of the MK3.6 model from the CMIP5 framework for three different greenhouse gas emission scenarios (RPC 2.6, 4.5, 8.5) of the Australian research agency CSIRO served as basis ([CDC, 2018](#); [Jeffrey et al., 2013](#)). A 25×25 km grid was laid over Europe and the weather data for a season from seeding to harvest were acquired for each coordinate. These simulations are provided on an averaged monthly basis and had to be downsampled to daily weather data which is required from DSSAT-Nwheat. The downsampling was conducted according to the following procedure (details in [appendix](#) section 1.4):

Table 1

The conducted experiments: The pure data-centric approach (DC) represents the most simple ML approach, where the field recordings are used solely as input features. The other approaches extend each other by increasing the feature and distribution space using simulations from the process-based model (DSSAT-Nwheat). The models were optimized using the MSE loss, except the loss function experiment (LF), which was trained to predict observed yield and simulated crop variables and conditions.

Experiment	Field recordings	Crop feature simulations	Synth. samples observed weather	Synth. samples historical weather	Synth. samples climate projections	Multiloss function
Data-centric (DC)	x	—	—	—	—	—
Synthetic features (SF)	x	x	—	—	—	—
Observed weather data (OWD)	x	x	x	—	—	—
Historical weather data (HWD)	x	x	x	x	—	—
Climate projections (CP)	x	x	x	x	x	—
Multiloss function (LF)	x	—	x	x	x	x

1. Calculate the standard deviation (σ) for each weather feature from the weather recordings of the farm sites in the observation period (2005–2021)
2. Sample synthetic daily weather values from the underlying distribution, given the mean (μ) from the monthly climate projections (CDC, 2018)

(a) Precipitation (P): $p \sim \text{LogNormal}(\mu, \sigma)$

(b) Temperature (T) (mean, max, min) and solar radiation (SR):

$$t \sim \text{Normal}(\mu, \sigma)$$

$$sr \sim \text{Normal}(\mu, \sigma)$$

(c) The following restrictions apply to keep the sampling of precipitation/temperature values as realistic as possible:

(d) i. $P = \max(0, x - 0.1)$

ii. $T_{\min} < T_{\text{mean}}$ and $T_{\text{mean}} < T_{\max}$

This resulted in three climate data sets for the different RPC levels (see [appendix](#) section 1.4 for details).

The synthetically generated samples were categorized by the weather data into different categories of distribution for later analysis. The data were divided into nine categories based on the features precipitation and temperature mean for the main vegetation period February–June. The subdivisions were based on the mean value and the standard deviation, where the lower values of the distributions were selected by the threshold $\mu - \sigma$, the upper values by the threshold $\mu + \sigma$ and the middle values in between. These categories were added individually to the field record data set and the model was retrained to check which data distribution produced a benefit for the approach.

3.9. Domain knowledge transfer via synthetic features and samples

Different approaches were applied for the hybrid ML approach to explore the most suitable strategy to transfer process knowledge from the DSSAT-Nwheat model into the ML models. The ‘synthetic feature’ (SF) approach aimed to integrate further knowledge about the crop status and growing conditions by simulating corresponding variables and appending them to the input feature vector (see [appendix](#) section 1.5). The approaches using observed weather data (OWD), historical weather data (HWD) and climate projections (CP) extended this concept by keeping the extended feature vector and increasing the training data set with generated synthetic yield samples. These samples were simulated by DSSAT-Nwheat with random soil and management as input features from the field recordings and weather data taken from the observed, historical, or climate projection weather data set. We assumed that the domain knowledge of the process-based model DSSAT-Nwheat is represented in the simulated yield value. For details of the data set constellations and the experiments we refer to [Fig. 1](#) and [Table 1](#).

3.10. Domain knowledge integration via multiloss

We conducted a multiloss experiment, where we extended the target feature vector and the loss function of the NN to learn the crop growth parameters in addition to the yield. This was inspired by the idea that the yield prediction benefits if the model learns the full underlying crop growth function of DSSAT-Nwheat and can generate the whole crop status at once. The crop parameters represent all crop variables from March–July that were simulated using DSSAT-Nwheat (see [appendix](#) section 2.3) and extended the target vector from $y^1 \rightarrow y^{127}$ to one yield and 126 crop parameters. For these experiments, we have deviated from the standard objective function (MSE) and have evaluated multiple loss function designs. The objective on which we chose the final loss function was still to achieve the best error in yield prediction. The most promising one was the ‘truncated’ loss ($L_{\text{truncated}}$) function approach, which is defined as:

Algorithm 1 Truncated loss function

$\hat{Y} \leftarrow \text{model}(X_{\text{train}})$

$l = \text{MSE}(\hat{Y}, Y)$ Calculate the MSE per target

$l_{\text{yield}} = l[\text{index}_{\text{yield}}]$

$l[l > l_{\text{yield}}] = l_{\text{yield}}$ Truncate loss values exceeding yield loss to the yield loss level

$$L_{\text{truncated}} = \frac{1}{127} \sum_{j=0}^{127} l_j$$

4. Results

4.1. Global model results

The benchmark models $E[x]$ and the DSSAT-Nwheat simulation did perform the poorest with average RMSE of 1.54 t/ha ($E[x]$, 18 % rRMSE, $-0.6 R^2$) and 2.2 t/ha (DSSAT-Nwheat, 59 % rRMSE, $-1.2 R^2$) and were outperformed by any other approach as shown in [Fig. 3](#). The RF outperforms the neural network for the pure data-centric and synthetic feature experiments but shows an incrementing error with every ongoing extension of the data set from 1.45 t/ha RMSE (17 % rRMSE, $0.07 R^2$) to 1.55 t/ha RMSE (18.1 % rRMSE, $-0.06 R^2$). Therefore, the RF was not considered in further experiments. In contrast, the NN did show steady improvement and the average RMSE decreased from 1.48 t/ha in the DC setting (17 % rRMSE, $0.3 R^2$) to 1.39 t/ha RMSE (16.2 % rRMSE, $0.15 R^2$), which refers to a drop of -8% . Compared to the $E[x]$ benchmark the RMSE decrease is -10.5% . Each consecutive experiment led to an average RMSE improvement of -1.6% , but the data set extensions led to a larger improvement (average of -1.8%) instead of the additional synthetic features (-1%). The CP approach tends to be the best model among all experiments. However, from a statistical point of view, most models are not significantly different from each other ([Fig. 3](#)). Over all 17 test years, one would expect the best model (CP) to be ranked at the 4.1th rank, followed by NN from the HWD experiment (4.4th) and the NN multiloss (4.6th). Across the years all the models showed variation in their performance quality. The CP model RMSE ranged from 1.07 t/hain 2005 to 1.64 t/hain 2012 (13 %–19 % rRMSE).

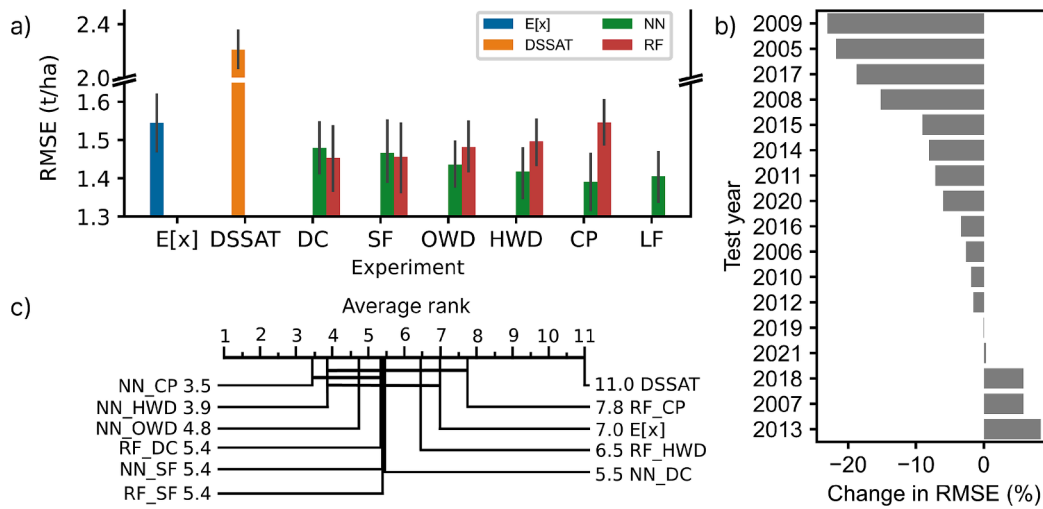


Fig. 3. a) The global averaged performance of the different approaches as described in Table 1 including the 95-confidence interval among the different test years. b) The RMSE change by test year of the CP model vs. the DC model and will be further investigated in Section 4.3. c) A critical difference diagram across all models, showing the average rank (ordered by ascending RMSE). Models that are not statistically significantly different ($p < 0.05$) are indicated with a bar.

Details of results per year are listed in appendix section 3.1.

Only four years exhibited worse prediction with the CP approach (Fig. 3 b). The multiloss experiment was the second best model, slightly performing worse than the CP experiment (1.4 t/haRMSE, 16.5 % rRMSE, 0.13 R^2). The other tested loss function did show a large performance gap and performed worse (appendix section 2.3). The standard deviation of the NN remained high, did not decrease and did not follow a trend ($\text{Var}(DC) = 0.02$, $\text{Var}(HWD) = 0.017$, $\text{Var}(CP) = 0.027$).

4.2. Multiloss results

The multiloss function performed 2nd best among the models, even though it was not fed with DSSAT-Nwheat-generated crop variables (see appendix section 2.3 for the additional targets in the output vector). The analysis of the rRMSE in Fig. 4 among the individual features showed

that a majority of the time-series features were learned properly, even though the RMSE remains in most situations higher than the yield loss.

Some features that represent crop status in March (or April) have no loss value as crop parameters have not been developed yet and were excluded from the rRMSE calculation. The heatmap in Fig. 4 shows a gradient over the months, whereas the error decreased closer to the harvest with a median loss decrease of 8.6 % per month. The crop conditions water and nitrogen stress remain features that were not appropriately learned at all with high errors peaking in rRMSE ≈ 250 %.

4.3. Effects of synthetic data

We found that the samples with high temperatures, generated at the upper part of the distribution tail, led to the greatest improvements in minimizing error (Fig. 5). In contrast, the precipitation of simulations

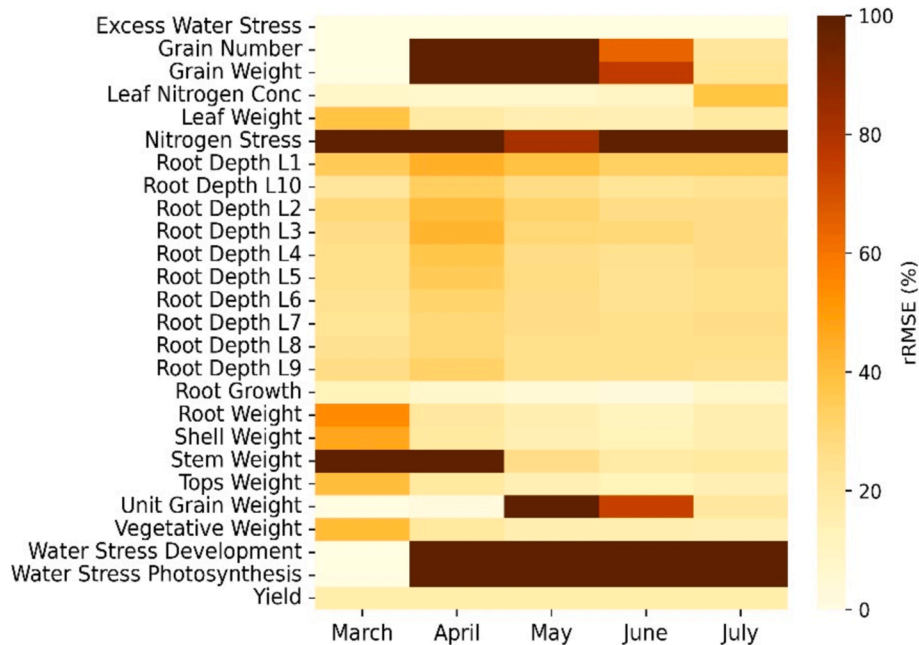


Fig. 4. rRMSE loss for the crop features over the time period of interest (only relative without unit, as most features have different units). The cutoff for the rRMSE was set at 100% for visualization purposes. The peaks in losses were for water stress development 283%, for water stress during photosynthesis 254%, and 241% for nitrogen stress. A gradient of the median rRMSE per month shows that states closer to harvest in July can be simulated better. The yield variable is not a time-series feature and is always related to the yield at the end of the season.

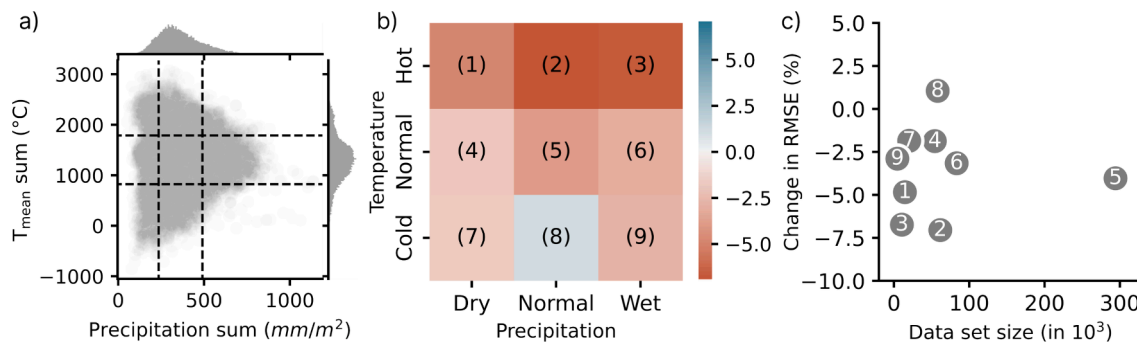


Fig. 5. a) Shows the 2D-distribution of the synthetic samples based on the defined thresholds for temperature mean and precipitation from February to June. This determines the assignment of the classes, that are displayed in b) and represents the change in RMSE when a category is attached to the field recordings. c) Represents the sample size of each category in relation to the change in error.

included in the training did not show consistent effects on performance. Overall, the best improvements were gained from subsets with data from hot-dry, hot-normal humid, and hot-wet conditions. Although the most extensive synthetic data set was sampled around the mean (category 5), it only appeared as the 4th rank regarding error reduction. Thus, the value of additional data appears more related to whether it enhances coverage of temperature extremes than to simply increasing data set size.

5. Discussion

We have demonstrated the enhancement of neural networks and random forest by process-based knowledge injection through the integration of the crop simulation model DSSAT-Nwheat. The expected value as the most simple predictor and the crop simulation model have been outperformed by our methods. The performance improvements with the inclusion of synthetic crop features in the feature vector aligns with findings from Feng et al. (2020), Feng et al. (2019), Corrales et al. (2022), Shahhosseini et al. (2021) and Paudel et al. (2021), who reported error decreases from 7 %–20 %. In contrast, the RF performance did not improve by any experiment involving synthetic samples. We hypothesize that the RF iteratively strengthens the learned pattern of the process-based model when the share of synthetic data in the training set increases. Thereby the error continuously increases, as DSSAT-Nwheat alone was the poorest predictor. However, in other ML and environmental domains, experiments have demonstrated the beneficial impact of synthetic data on RF (Saha et al., 2021). Similar advantages were observed for NN using synthetic data in environmental-related research (Liu et al., 2024) and computer vision for object detection/segmentation (Fabbri et al., 2021). We assume that the NNs contain a greater capacity to learn overarching, general crop growth rules embedded in the synthetic data while still acquiring specific patterns from the field recordings. Similar results were reported for field data experiments (Kallenberg et al., 2023) or solely synthetic data sets (Pylianidis et al., 2022). The final level of error of the NN ranging from 13 % – 19 % rRMSE aligns with the error range reported by others using hybrid ML approaches (Feng et al., 2020; Shahhosseini et al., 2021; Paudel et al., 2021) and pure data-driven experiments (Wang et al., 2020; von Bloh et al., 2023). Experiments on a national/subnational scale showed in general lower errors due to reduced variability in the system than field level predictions. We emphasize the temporal, spatial, and agronomic variability of our data set, which was higher than any other field level data set. Shahhosseini et al. (2021) mentions that climate extreme years did strongly benefit from a hybrid approach, but we can only partially validate these findings. While years with the most improvements through the CP approach exhibit distinct climatic characteristics, among the few years that suffered at least one year was subject to one of the most severe drought situations in the past decade (2018). In our experiments, we report significant gains through the generation of

temperature extremes at the upper tail of the distribution. Dry, humid, and wet conditions also lead to a reduction in error, but mainly when experienced in combination with hot temperatures. These findings are supported by Bracho-Mujica et al. (2024), Lobell & Asseng (2017) and Asseng et al. (2015), who state a large impact on wheat yield due to temperature increase and climate change.

Our results show that generating synthetic data based on climate projections increases the seasonal climate variability in a training dataset which enhances the robustness of ML models and reduces the model error. This model improvement applies both to past and future climate predictions of cropping systems and suggests, that data from a wider range of climates are more effective in training ML models than from a less climate variable background. We expect this information to be particularly valuable when reproducing the approach for data generation, as it allows for the more efficient allocation of input resources such as computing time, computing power, and storage costs.

The multiloss function was successfully applied and shows the potential to do without DSSAT-Nwheat in the future when other crop parameters should be predicted alongside yield. Crop growth processes (e. g. development of biomass, leaves, roots, etc.) are learned well, in contrast to crop conditions like drought and nitrogen stress. We believe this can be attributed to the presence of more signals in the time-series features for later time points. Shahhosseini et al. (2021) states drought-related conditions are the most important to gain improvements for hybrid ML, but we conclude they are hard to learn. Here, the suitability of the truncated approach may be reconsidered to determine its appropriateness, given that it may constrain the loss values of poorly learned features too much. However, this was required to achieve the most accurate yield prediction. Other loss functions might be more suitable for the objective of learning the DSSAT-Nwheat decision rules without a specific focus.

Key factors for successful knowledge transfer include the availability of comprehensive field data for machine learning and the quality of the process-based model. Major crops like corn or soybean have an extensive research history in data acquisition and process-model development, enabling robust knowledge transfer to ML models. In contrast, niche crops or fruits like strawberries or spinach are less likely to benefit from this approach with crop specific ML models. If a well-researched process model and sufficient field records are available, our knowledge transfer method applies to any crop or region, but the absence or limitation of either component restricts the transferability and might leave process-based models as the more favorable solution. Ideas as to which crops, spatial scale or region our approach could be applied to can be found in Clark et al. (2023).

We propose the research of generative models to produce more realistic fine-gridded temporal weather data or satellite images to incorporate them as an additional feature source (Lobell et al., 2020). Generative approaches like diffusion or foundation models might be worth developing or fine-tuning for these use cases. Identifying samples

and situations that did not benefit from the knowledge-informed approaches would be another important next step to evaluate approaches how these situations could also be improved by the informed ML.

6. Conclusion

Farmers, burdened with growing responsibilities for sustainable practices, require detailed field-level yield predictions to quantify the impact of external and internal factors and proactively address challenges. Accurate yield monitoring is crucial for optimizing resources and mitigating risks, especially with a rising world population demanding increased nutrient supply. Process-based models tend to overgeneralize and are unsuitable for field-specific applications. Our results indicate that in data-scarce domains and where big data is not foreseeable, integration with process-based models might be crucial to make ML models robust for real-world applications. Here we presented that synthetic samples simulated from features at the tails of the distribution can gain the largest improvements.

We demonstrated that not the size but the diversity of the synthetic data matters that leads to yield prediction performance gains. Adapting the multiloss to learn process rules in parallel to the objective task might be a way to softly guide the learning process via the loss function when hard constraining (e.g., via mathematical equations as in physics or robotics) is not easily feasible. The successful approach of domain knowledge injection shows that agricultural sectors can benefit from a combination of process-based and ML models. Future research on the integration of core biophysical equations or human expert knowledge might be promising to investigate to boost hybrid machine learning applications.

Data availability statement

Data supporting our findings are available at <https://gitlab.com/cha-ir-of-digital-agriculture/hybridml-yieldprediction-data-store> with restrictions due to data protection agreements. Climate datasets for DSSAT simulations, climate projections, and genotype files are provided in the repository. All data are made available under a CC BY-NC 4.0 license.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT v3.5 in order to edit and improve spelling, language and readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRedit authorship contribution statement

Malte von Bloh: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David Lobell:** Writing – review & editing, Validation, Supervision, Software, Methodology, Investigation. **Senthold Asseng:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2024.109606>.

Data availability

Please see the data availability statement. Data have been made public in gitlab

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